Zoie Telkamp

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Education

PhD in Astronomy Expected 2027
Master of Science in Astronomy 2023
University of Virginia
Advisor: Adele Plunkett
Dissertation: "A Multi-Scale Investigation of Star Formation Across the Mass Spectrum"

Bachelor of Arts in Astrophysics 2020
University of California, Berkeley
Advisor: Gaspard Duchêne

Research Experience

Graduate Dissertation Research
University of Virginia

Investigating Protostellar Outflows with JWST NIRCam 2025–Present
Complex Organic Molecules in Protostars with ALMA Spectral Surveys (COMPASS) Team

- Reduced JWST NIRCam images and characterized >30 protostellar outflows in eight star-forming regions; now comparing outflow orientations with magnetic field directions from archival dust polarization data.
- Building an interactive tool for viewing ALMA and JWST spectra for the COMPASS website.

An Archival ALMA Analysis of Disks in Massive Star-Forming Regions 2023–2026
Advisors/Mentors: A. Plunkett, J. C. Tan

- Identified and characterized >100 candidate protostellar disks using high-resolution archival ALMA continuum images of 17 massive star-forming regions.
- Compared disk orientations to literature outflow orientations and performed complementary IR spectral energy distribution (SED) fitting to test massive star formation model predictions (Telkamp et al. 2026, submitted).

The SOFIA Massive (SOMA) Star Formation Survey 2021–2025
Advisors/Mentors: R. Fedriani, J. C. Tan

- Characterized 34 protostars in crowded Galactic star-forming regions via SOFIA, Spitzer, and Herschel photometry and SED fitting, nearly doubling the size of the survey.
- Added empirical constraints and ran restricted SED fits of SOMA Paper I–V protostars to test the environmental dependence of massive star formation (Telkamp et al. 2025).
- Led development of version 2.0 of the open-source Python package `sedcreator` (Fedriani et al. 2023, Telkamp et al. 2025; used in 15+ publications), adding tools for photometry and SED fitting in crowded star-forming regions.

Additional Graduate Research
NSF-Simons AI Institute for Cosmic Origins (CosmicAI)

Benchmarking Large Language Models for Astronomical Data Access Summer 2026
Advisors/Mentors: A. Plunkett, B. Mason, S. Offner, S. Juneau, R. Nikutta, R. Loomis

- Built `AstroDataBench`, an open-source benchmarking framework for evaluating LLM reliability in facilitating archival astronomy data access, in collaboration with NRAO and NOIRLab astronomers (currently being expanded by a CosmicAI intern).

Undergraduate and Post-Baccalaureate Research

University of California, Berkeley

Machine Learning Analysis of Edge-On Protoplanetary Disks 2018–2021

Advisor: G. Duchêne

- Led a project developing a machine learning framework (autoencoder and random forest regression) to generate synthetic edge-on protoplanetary disk images 1000× faster than radiative transfer calculations, to facilitate efficient disk modeling (Telkamp et al. 2022).
- Developed morphological analysis methods for HST protoplanetary disk images (Villenave et al. 2020) and participated in observing runs at Keck and Lick observatories.

Awards and Recognition

Jefferson Scholars Foundation Dissertation Year Fellowship 2026–2027

Emma Williams Prize in Astronomy (University of Virginia) 2025

American Astronomical Society (AAS) Nova, "Selections from 2025" Feature 2025

Telkamp et al. 2025 featured as one of the most-downloaded AAS journal articles published in 2025

Virginia Space Grant Consortium (VSGC) Graduate STEM Research Fellowship 2023–2025

International Astronomical Union Grant (for Kavli-IAU S383) 2023

Dorothea Klumpke Roberts Prize in Astronomy (UC Berkeley) 2020

Undergraduate Research Apprentice Program Summer Award (UC Berkeley) 2020

Teaching and Mentorship Experience

Graduate Student Teaching Assistant 2022–2026

University of Virginia

- **Graduate:** Computational Astrophysics; Astronomical Techniques
- **Undergraduate:** Introduction to Astrophysics; Stars, Galaxies, and the Universe; Introduction to the Sky and Solar System; Origins of Almost Everything; Black Holes

Summer Student Co-Supervisor Summer 2022

Chalmers/Virginia Initiative on Cosmic Origins (CICO/VICO); with R. Fedriani and J. C. Tan

- Samuel Crowe (University of Virginia): "Massive Protostars in the Galactic Center," presented at the 241st AAS Meeting (talk) and the CICO Summer 2022 Workshop.
- Alejandro Romar Tejeiro (Universidad Nacional de Educación a Distancia): "Infrared Light in the Dark—Hunting for Protostars in IRDCs," presented at the CICO Summer 2022 Workshop.

Publications

First Author

Telkamp, Z., Plunkett, A., Tan, J. C., 2026, "An Archival ALMA Survey of Disks in Massive Star-Forming Regions," submitted to ApJ; revisions in progress

Telkamp, Z., Fedriani, R., Tan, J. C., Law, C. Y., Zhang, Y., Plunkett, A. (+13 co-authors), 2025, "The SOFIA Massive (SOMA) Star Formation Survey. V. Clustered Protostars," ApJ, 986, 15

Telkamp, Z., Martínez-Palomera, J., Duchêne, G. (+4 co-authors), 2022, "A machine learning framework to predict images of edge-on protoplanetary disks," ApJ, 939, 73

Contributing Author

Plunkett, A., et al. (incl. **Telkamp, Z.**; 28 authors), 2026, "Complex Organic Molecules in Protostars with ALMA Spectral Surveys (COMPASS). II. Approach to data reduction and products for line-rich broadband (sub-)millimeter spectra," A&A, in press

Coutens, A., et al. (incl. **Telkamp, Z.**; 28 authors), 2026, "Complex Organic Molecules in Protostars with ALMA Spectral Surveys (COMPASS). III. CH_3OH isotopic fractionation in the low-mass protostar BHR71-IRS1," A&A, in press

Gigli, D., Gorai, P., Law, C. Y., Tan, J. C., Bonfand, M., Rahman, T., Zhang, Y., Taniguchi, K., Fedriani, R., **Telkamp, Z.** (+2 co-authors), 2026, "The SOMA MM Survey. I. An Astrochemical Census of Massive Protostars," A&A, in press

Jørgensen, J., Coutens, A., Drozdovskaya, M., Lee, J.-E., Plunkett, A., Maret, S., Belloche, A., Harsono, D., **Telkamp, Z.** (+20 co-authors), 2026, "Complex Organic Molecules in Protostars with ALMA Spectral Surveys (COMPASS). I. Overview of the ALMA Large Program," A&A, 714, A49

Oakey, P., Yang, Y. L., Tan, J. C., Bisbas, T., Fedriani, R., Tanaka, K., **Telkamp, Z.** (+3 co-authors), 2026, "The SOMA Atomic Outflow Survey. I. An Atomic OI and Highly Ionized OIII Outflow from Massive Protostar G11.94-00.62," ApJ, 1003, 26

Crowe, S., Fedriani, R., Tan, J. C., Whittle, M., Zhang, Y., Caratti o Garatti, A., Farias, J. P., Gautam, A., **Telkamp, Z.** (+24 co-authors), 2024, "Near-infrared observations of outflows and young stellar objects in the massive star-forming region AFGL 5180," A&A, 682, A27

Angelo, I., Duchêne, G., Stapelfeldt, K., **Telkamp, Z.** (+8 co-authors), 2023, "Demographics of Protoplanetary Disks: A Simulated Population of Edge-on Systems," ApJ, 945, 130

Fedriani, R., Tan, J. C., **Telkamp, Z.** (+13 co-authors), 2023, "The SOFIA Massive (SOMA) Star Formation Survey. IV. Isolated Protostars," ApJ, 942, 7

Finn, M., Mazzei, R., Drechsler, B., **Telkamp, Z.** (+2 co-authors), 2023, "How Informal Science Education Influences Elementary Students' Perceptions of Science and Themselves," J. STEM Outreach, 6, 1

Villenave, M., Ménard, F., Dent, W., Duchêne, G., Stapelfeldt, K., Benisty, M., Boehler, Y., Plas, G., Pinte, C., **Telkamp, Z.** (+8 co-authors), 2020, "Observations of edge-on protoplanetary disks with ALMA. I. Results from continuum data," A&A, 642, A164

Talks

Infrared Processing and Analysis Center (IPAC) Lunch Seminar 2026
How Do Massive Stars Form? A Multi-Scale View of Protostars and Their Environments

Virginia Tech Astronomy Seminar (Invited) 2026
A Multi-Scale Comparison of Star Formation from Low to High Masses

Jefferson Scholars Foundation Fellows Symposium 2026
How Do Massive Stars Form? A Multi-Scale View of Protostars and Their Environments

NSF-Simons AI Institute for Cosmic Origins Webinar 2026
Benchmarking LLM Use in Multiwavelength Astronomy

Chalmers/Virginia Initiative on Cosmic Origins Spring Workshop 2026
An Archival ALMA Survey of Disks Around Massive Protostars

Virginia Space Grant Consortium Graduate Research Conference 2025
Investigating Massive Star Formation with ALMA

Bob Rood Memorial Symposium 2025
The SOFIA Massive (SOMA) Star Formation Survey: Clustered Regions

Bob Rood Memorial Symposium 2024
Probing Massive Star Formation with the ALMA Archive

National Radio Astronomy Observatory "TUNA" Talk (Invited) 2023
Zooming into Massive Star Formation: A Multi-Scale Approach

Our Galactic Ecosystem: Opportunities & Diagnostics in the IR and Beyond 2022
The SOMA Survey: Probing Massive Star Formation Across Galactic Environments

Chalmers/Virginia Initiative on Cosmic Origins Spring Workshop 2021
The SOFIA Massive (SOMA) Star Formation Survey: New Algorithms and New Environments

237th Meeting of the American Astronomical Society 2021
A Physics-Aware Neural Network to Generate Edge-On Protoplanetary Disk Images

Posters

DISks in COntext (DISCO), National Radio Astronomy Observatory 2026
An Archival ALMA Survey of Candidate Disks in Massive Star-Forming Regions

2nd Annual CosmicAI Science Conference 2026
Benchmarking LLM Use in Multiwavelength Astronomy

ALMA at 10 years: Past, Present, and Future 2023
Probing Massive Star Formation with the ALMA Archive

From Stars to Galaxies II, Chalmers University of Technology 2022
The SOMA Survey: Probing Massive Star Formation Across Galactic Environments

Open-Source Python Packages

`sedcreator` *(Co-Lead Developer; with R. Fedriani)* Documentation | GitHub
Photometry and SED fitting for protostars in isolated or crowded environments

`AstroDataBench` *(Lead Developer)* Article | GitHub
Benchmark for evaluating LLMs on multiwavelength data discovery and domain questions

Leadership and Outreach

Graduate School of Arts and Sciences Council 2023–2024
Representative for the UVA Astronomy Department

Dark Skies, Bright Kids (DSBK) Assessments 2021–2023
Researcher and co-author of a publication in the Journal of STEM Outreach investigating the enrichment of science education in under-served school systems (Finn et al. 2023)

Additional Skills

Programming: Python, ADQL, UNIX, JavaScript, HTML

Archives/Observatories: MAST (JWST, HST), IRSA (Spitzer, Herschel, SOFIA), ALMA, JCMT

Software and Packages: JWST Science Calibration Pipeline, STPSF/WebbPSF, Astroquery, Astropy, Astrodendro, Photutils, Scikit-learn, CASA, Docker, Harbor, Pytest, LaTeX, Git, DS9

AI/Machine Learning: Large Language Models (LLMs), Deep Neural Networks, Principal Component Analysis (PCA), Random Forest Regression